

5	(a) (i)	$v = f\lambda$	C1	
		$\lambda = 40 / 50 = 0.8(0) \text{ m}$	A1	[2]
	(ii)	waves (travel along string and) reflect at Q / wall / fixed end	B1	
		incident and reflected waves interfere / superpose	B1	[2]
	(b) (i)	nodes labelled at P, Q and the two points at zero displacement	B1	
		antinodes labelled at the three points of maximum displacement	B1	[2]
	(ii)	$(1.5\lambda \text{ for PQ hence } PQ = 0.8 \times 1.5) = 1.2 \text{ m}$	A1	[1]
	(iii)	$T = 1 / f = 1/50 = 20 \text{ ms}$	C1	
		5 ms is $\frac{1}{4}$ of cycle	A1	
		horizontal line through PQ drawn on Fig. 5.2	B1	[3]